

# Fundamentals of Linear Control:

## A Concise Approach

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**Chapter 6: Controller Design**

[linearcontrol.info/fundamentals](http://linearcontrol.info/fundamentals)

# 6.1 Second-Order Systems

## Characteristic equation

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad \omega_n > 0$$

- $\zeta$ : damping ratio
- $\omega_n$ : natural frequency

## Roots

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s - s_1)(s - s_2), \quad s_1 = -\omega_n(\zeta + \sqrt{\zeta^2 - 1}), \quad s_2 = -\omega_n(\zeta - \sqrt{\zeta^2 - 1})$$

## Real roots

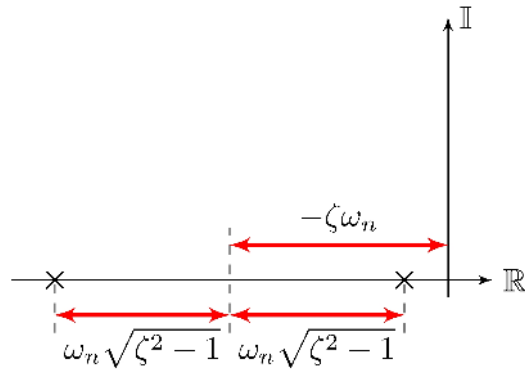
$$(|\zeta| \geq 1)$$

## Complex roots

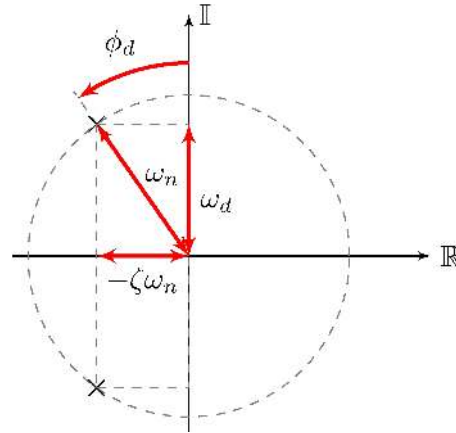
$$(|\zeta| < 1)$$

# Second-Order Systems

real roots,  $(|\zeta| \geq 1)$



complex roots,  $(|\zeta| < 1)$



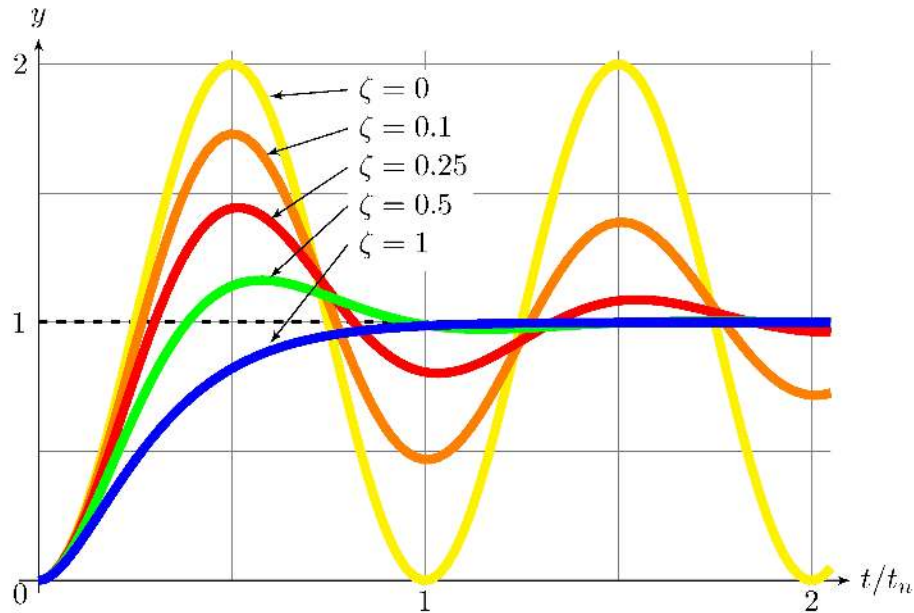
## Roots

$$\begin{aligned} s &= -\omega_n (\zeta \pm \sqrt{\zeta^2 - 1}), & \omega_d &= \omega_n \sqrt{1 - \zeta^2}, & \\ \phi_d &= \sin^{-1} \frac{\zeta \omega_n}{\omega_n} = \tan^{-1} \frac{\zeta}{\sqrt{1 - \zeta^2}} = \sin^{-1} \zeta \end{aligned}$$

## Asymptotic stability

$$(\zeta > 0)$$

# Second-Order Systems

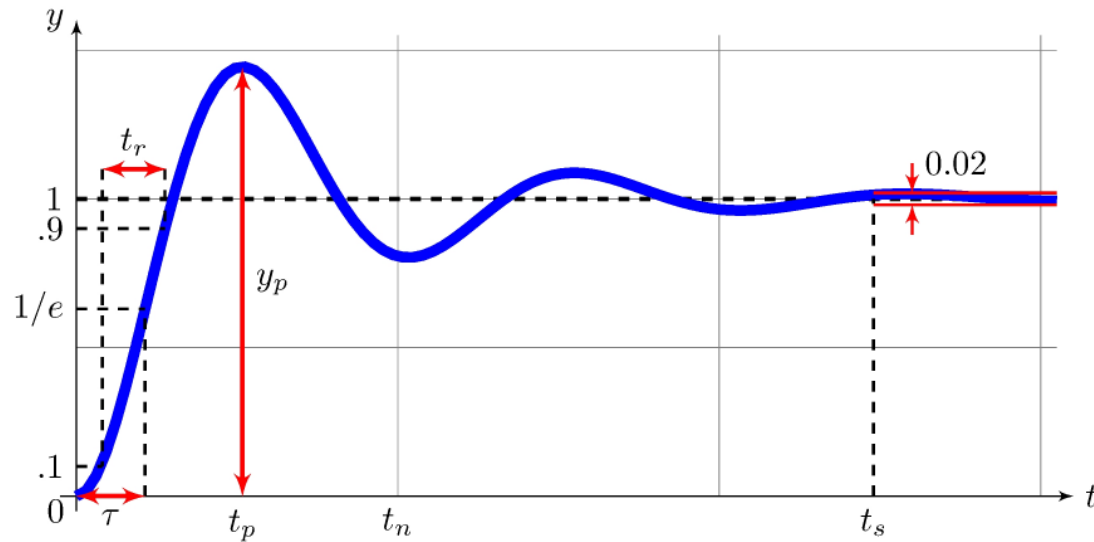


## Step response

$$\begin{aligned} G(s) &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, & y(t) &= 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \pi/2 - \phi_d), \quad t \geq 0 \end{aligned}$$

# Second-Order Systems

## Step response

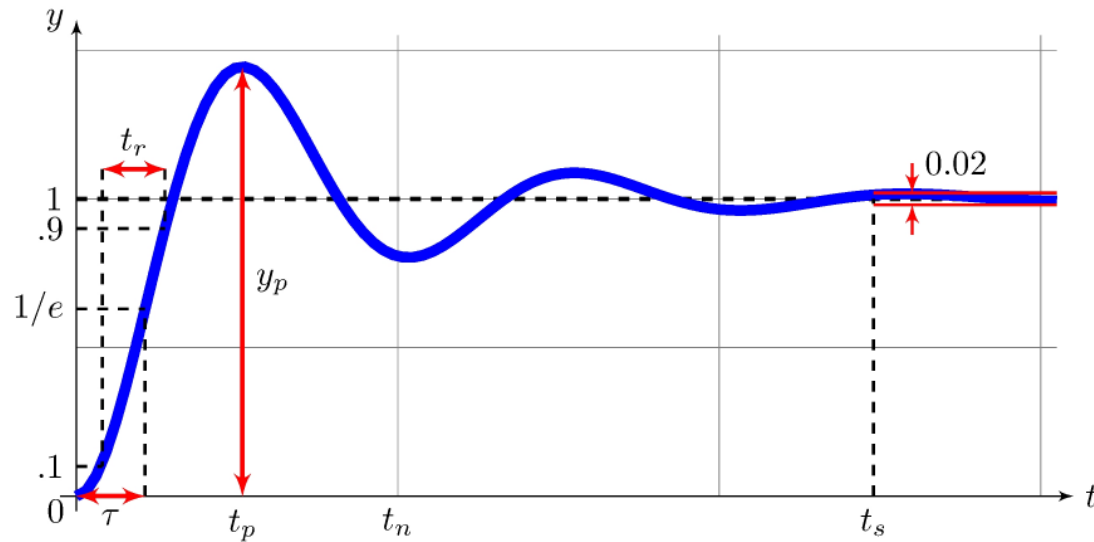


## Time-constant and Rise-time

$$\begin{aligned} \frac{t_r}{t_n} &\approx 0.16 + 0.14 \zeta + 0.24 \zeta^3, & \frac{\tau}{t_n} &\approx 0.19 + 0.1 \zeta + 0.054 \zeta^3, & t_n &= \frac{2\pi}{\omega_n} \end{aligned}$$

# Second-Order Systems

## Step response



## Overshoot and Settling-time

$$\begin{aligned} t_p &= \frac{\pi}{\omega_d}, & y_p &= 1 + e^{-\left(\frac{\zeta \pi}{\sqrt{1-\zeta^2}}\right)}, & \\ t_s &= \frac{\log(50)}{\zeta \omega_n} \approx \frac{3.9}{\zeta \omega_n} \end{aligned}$$

# Example: car model with integral feedback

## Open-loop

$$\begin{aligned} G(s) &= \frac{\frac{p}{m}}{s + \frac{b}{m}}, & K(s) &= \frac{K_i}{s} \end{aligned}$$

## Closed-loop

$$\begin{aligned} H(s) &= \frac{K(s) G(s)}{1 + K(s) G(s)} = \frac{\frac{p}{m} K_i}{s^2 + \frac{b}{m} s + \frac{p}{m} K_i} \end{aligned}$$

## Second-order

$$\begin{aligned} \omega_n &= \sqrt{\frac{p}{m} K_i}, & \zeta &= \frac{b/m}{2 \sqrt{(p/m) K_i}} = \frac{b}{2 \sqrt{p m K_i}} \end{aligned}$$

# Example: car model with proportional-integral feedback

## Open-loop

$$\begin{aligned} G(s) &= \frac{\frac{p}{m}}{s + \frac{b}{m}}, & K(s) &= K_p + \frac{K_i}{s} \end{aligned}$$

## Closed-loop

$$\begin{aligned} H(s) &= \frac{K(s) G(s)}{1 + K(s) G(s)} = \frac{\frac{p}{m} (K_p s + K_i)}{s^2 + (\frac{b}{m} + \frac{p}{m} K_p) s + \frac{p}{m} K_i} \end{aligned}$$

## Second-order

$$\begin{aligned} \omega_n &= \sqrt{\frac{p}{m} K_i}, & \zeta &= \frac{b/m + p/m K_p}{2 \sqrt{(p/m) K_i}} = \frac{b + p K_p}{2 \sqrt{p m K_i}} \end{aligned}$$



## Example: simple pendulum

### Open-loop (stable equilibrium)

$$G_0(s) = \frac{1/J_r}{s^2 + (b/J_r)s + m\ell g/r}$$

### Second-order

$$\omega_{n,\mathrm{ol}} = \sqrt{\frac{m\ell g}{J_r}}, \quad \zeta_{\mathrm{ol}} = \frac{b}{2\sqrt{J_r m\ell g}}$$

### Open-loop (unstable equilibrium)

$$G_{\pi}(s) = \frac{1/J_r}{s^2 + (b/J_r)s - m\ell g/r}$$

# Example: simple pendulum

## Open-loop

$$G_{\pi}(s) = \frac{1}{J_r s^2 + b s - m \ell g}$$

## Proportional feedback (unstable equilibrium)

$$H_{\pi}(s) = \frac{K G_{\pi}(s)}{1 + K G_{\pi}(s)} = \frac{K}{J_r s^2 + b s + K - m \ell g}$$
$$\omega_{n\pi} = \sqrt{\frac{K - m \ell g}{J_r}}, \quad \zeta_{\pi} = \frac{b}{2 \sqrt{J_r (K - m \ell g)}}$$

## Closed-loop stability

$$(K > m \ell g), \text{ for example } K = 3 m \ell g \implies \omega_{n\pi} = \sqrt{2} \omega_{n\mathrm{ol}}, \quad \zeta_{\pi} = \frac{1}{\sqrt{2}} \zeta_{\mathrm{ol}}$$

# Example: simple pendulum

## Open-loop

$$\begin{aligned} G_0(s) &= \frac{1}{J_r s^2 + b s + m \, g \, r}, & \omega_{n_{\mathrm{ol}}} &= \sqrt{\frac{m \, g \, r}{J_r}}, \\ & & \zeta_{\mathrm{ol}} &= \frac{b}{2 \sqrt{J_r m \, g \, r}} \end{aligned}$$

## Proportional feedback (stable equilibrium)

$$\begin{aligned} H_0(s) &= \frac{K G_0(s)}{1 + K G_0(s)} = \frac{K}{J_r s^2 + b s + K + m \, g \, r}, & \omega_{n_0} &= \sqrt{\frac{K + m \, g \, r}{J_r}}, \\ & & \zeta_0 &= \frac{b}{2 \sqrt{J_r (K + m \, g \, r)}} \end{aligned}$$

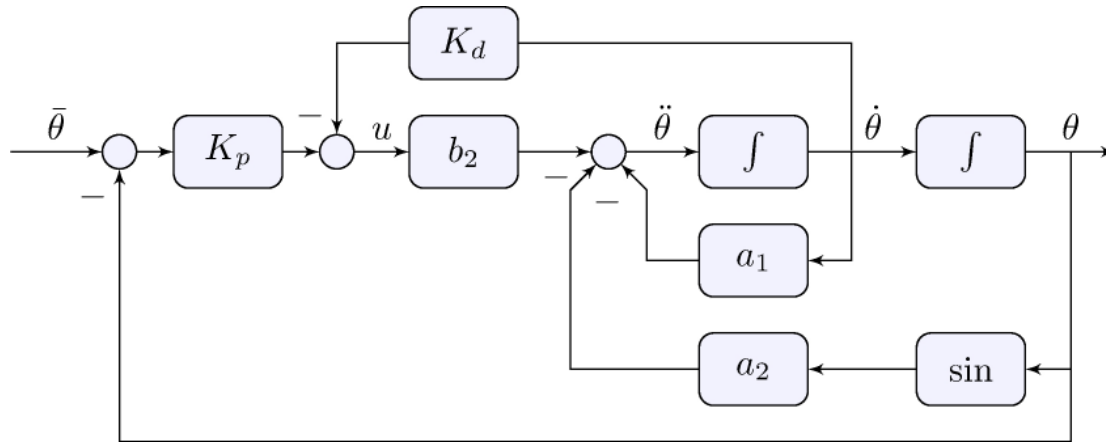
## Closed-loop stability

$$(K > -m \, g \, r), \text{ for example } \begin{aligned} K &= 3 \, m \, g \, r & \implies & \omega_{n_0} = 2 \, \omega_{n_{\mathrm{ol}}}, \\ & & \zeta_0 &= \frac{1}{2} \zeta_{\mathrm{ol}} \end{aligned}$$

## 6.2 Derivative Action

### Example: simple pendulum

$$u(t) = K_p (\bar{\theta}(t) - \theta(t)) - K_d \dot{\theta}(t)$$

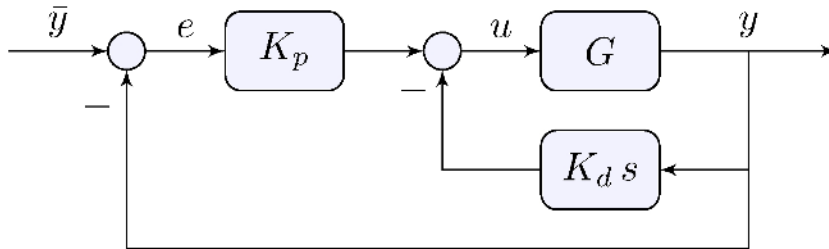


# Proportional-derivative feedback

$$u(t) = K_p (\bar{y}(t) - y(t)) - K_d \dot{y}(t)$$

## Frequency-domain

$$U(s) = K_p E(s) - K_d s Y(s), \quad E(s) = \bar{Y}(s) - Y(s)$$

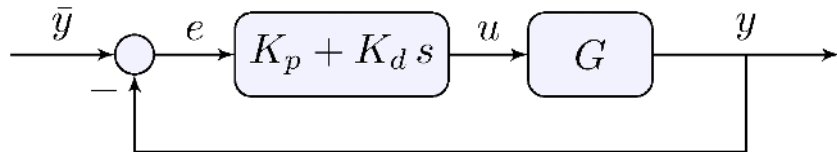


# Proportional-derivative feedback

$$\begin{aligned} u(t) &= K_p e(t) + K_d \dot{e}(t), & e(t) &= \bar{y}(t) - y(t) \end{aligned}$$

## Frequency-domain

$$\begin{aligned} U(s) &= (K_p + s K_d) E(s), & E(s) &= \bar{Y}(s) - Y(s) \end{aligned}$$



- Same if  $\bar{y}$  is constant!
- If not, extra zero

# Example: simple pendulum

## Unstable equilibrium

$$\begin{aligned} H_{\pi}(s) &= \frac{(K_p + K_d s) G_{\pi}(s)}{1 + (K_p + s K_d) G_{\pi}(s)} = \frac{K_p + K_d s}{J_r s^2 + (K_d + b) s + (K_p - m \ell g \ell, r)}, \\ \omega_{n_{\pi}} &= \sqrt{\frac{K_p - m \ell g \ell, r}{J_r}}, \quad \zeta_{\pi} = \frac{K_d + b}{2 \sqrt{J_r (K_p - m \ell g \ell, r)}} \end{aligned}$$

- Choose  $(K_p)$  to set desired natural frequency, e.g.  $K = 3 m g r \implies \omega_{n_{\pi}} = \sqrt{2} \ell, \omega_{n_{\text{ol}}}$
- Choose  $(K_d)$  to set desired damping ratio, e.g.  $\zeta_{\pi} = \frac{\sqrt{2}}{2} \approx 0.7 \implies K_d = 2 \ell, \zeta_{\pi} \sqrt{J_r (K_p - m \ell g \ell, r)} - b = 2 \sqrt{J_r m \ell g \ell, r} - b$
- Similarly around stable equilibrium

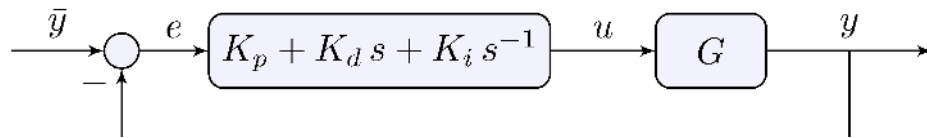
# 6.3 Proportional-Integral-Derivative Control

## Proportional-integral-derivative feedback

$$u(t) = K_p e(t) + K_d \dot{e}(t) + K_i \int_0^t e(\tau) d\tau, \quad e(t) = \bar{y}(t) - y(t)$$

## Frequency-domain

$$U(s) = (K_p + K_d s + K_i s^{-1}) E(s) = \frac{K_d s^2 + K_p s + K_i}{s} E(s), \quad E(s) = \bar{Y}(s) - Y(s)$$



- integral action: asymptotic tracking of constant reference
- derivative action: better “damping”
- Extensive literature on properties of PID control

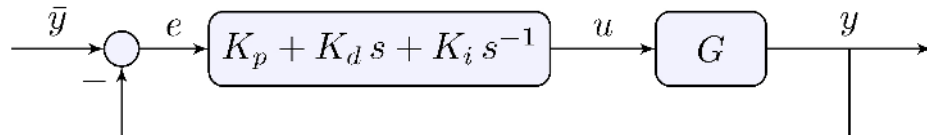


# Proportional-integral-derivative feedback

$$u(t) = K_p e(t) + K_d \dot{e}(t) + K_i \int_0^t e(\tau) d\tau, \quad e(t) = \bar{y}(t) - y(t)$$

## Frequency-domain

$$U(s) = (K_p + K_d s + K_i s^{-1}) E(s) = \frac{K_d s^2 + K_p s + K_i}{s} E(s), \quad E(s) = \bar{Y}(s) - Y(s)$$

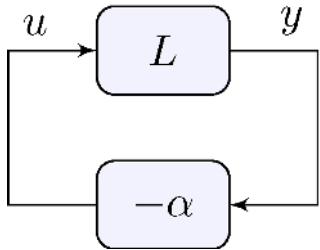


- Transfer-function is not proper: not realizable!
- If  $(G)$  is second-order then closed-loop transfer-functions are third-order
- Calculation of the characteristic roots is *already* too complicated

# 6.4 Root-Locus

## The basic idea

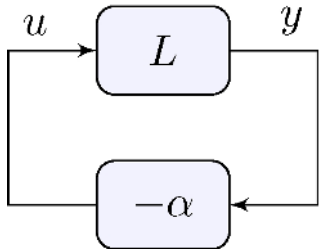
- Can we quickly *sketch* the location of the closed-loop poles without calculating them?
- Root *locus* (=) location of the roots
- Set of simple rules for sketching the root-locus of the feedback system



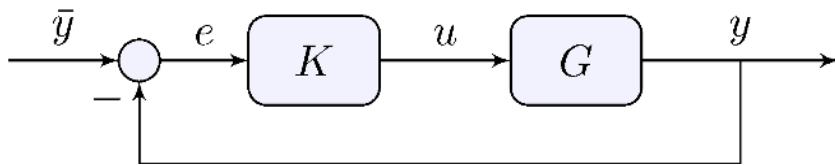
as a function of  $\alpha > 0$

- Make your feedback system look like it and apply the rules
- $L$  is the *loop transfer-function*

# Feedback diagram for root-locus



## Example



## Closed-loop poles

$$\begin{aligned} 1 + \alpha L(s) &= 0, & K(s) &= \alpha C(s), & L(s) &= C(s) G(s) \end{aligned}$$

# Root-locus rules

## Key observation

$$\left[ \begin{array}{l} \text{label{eq:rlcrossing}} \end{array} \right. L(s) \approx -\frac{1}{\alpha} < 0 \implies \angle L(s) \approx \pi \end{array} \right]$$

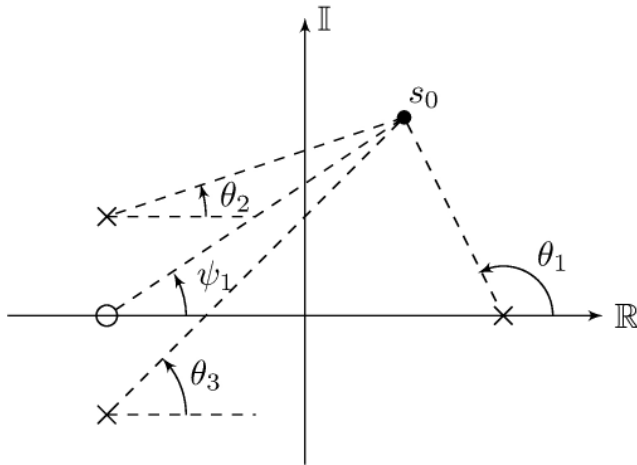
## Rational loop transfer-function

$$\left[ \begin{array}{l} L \approx \beta \frac{N_L}{D_L} = \frac{\beta}{(s - z_1)(s - z_2) \cdots (s - z_m)} \frac{(s - p_1)(s - p_2) \cdots (s - p_n)}{1} \end{array} \right] \text{ then } \left[ \begin{array}{l} \angle L = \angle \beta + \angle N_L - \angle D_L = \angle \beta + \sum_{i=1}^m \angle (s - z_i) - \sum_{k=1}^n \angle (s - p_k) \end{array} \right]$$

# Root-locus rules

## Phase can be calculated graphically

$$\begin{aligned} \angle L(s_0) &= \angle \beta + \sum_{i=1}^m \angle (s_0 - z_i) - \sum_{k=1}^n \angle (s_0 - p_k) \\ &= \angle \beta + \psi_1 - \theta_1 - \theta_2 - \theta_3 \end{aligned}$$



# Root-locus rules

## Assumption

$G(s)$  rational and strictly proper with real coefficients

## Rules

- Root-locus is a set of  $n$  continuous curves
- The curves:
  - begin at the poles of  $G(s)$  
$$\lim_{\alpha \rightarrow 0} \alpha D_L(s) \left( 1 + \alpha L(s) \right) = \lim_{\alpha \rightarrow 0} D_L(s) + \alpha N_L(s) = D_L(s)$$
  - end at the zeros of  $G(s)$  
$$\lim_{\alpha \rightarrow \infty} \alpha^{-1} D_L(s) \left( 1 + \alpha L(s) \right) = \lim_{\alpha \rightarrow \infty} \alpha^{-1} D_L(s) + N_L(s) = N_L(s)$$
- $(n - m \geq 1)$  curves go to infinity

# Root-locus rules

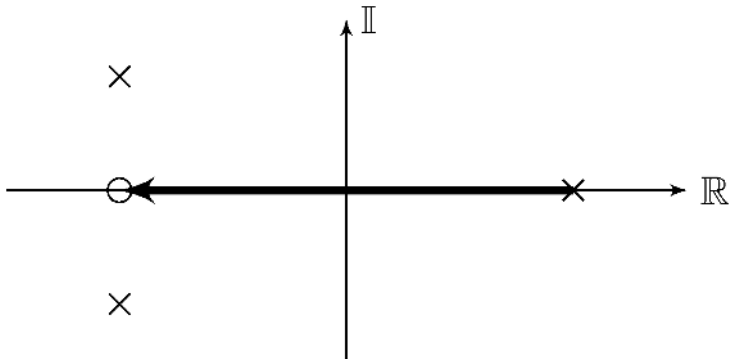
## Assumption

$\mathcal{L}(s)$  rational and strictly proper with real coefficients and  $\beta > 0$

## Rule

- $s \in \mathbb{R}$  is in the root-locus iff  $\mathcal{L}(s)$  has an odd number of poles and zeros with real part greater than or equal to  $s$

## Example (complex locus not shown)



- What would the  $\beta < 0$  version be?

# Root-locus rules

## Assumption

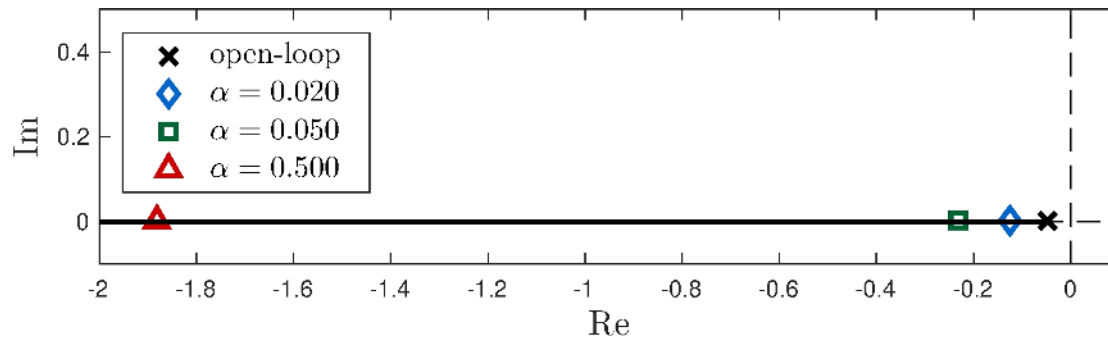
$L(s)$  rational and strictly proper with real coefficients and  $\beta > 0$

## Rule

- $s \in \mathbb{R}$  is in the root-locus iff  $L(s)$  has an odd number of poles and zeros with real part greater than or equal to  $s$

## Example: car model under proportional feedback

$$\alpha = K_p, \quad L(s) = \frac{p}{s + b} = \frac{3.7}{s + 0.05}$$





# Root-locus rules

## Assumption

$(L)$  rational and strictly proper with real coefficients

## Rule

- If  $(n > m + 1)$ , then  $(n - m)$  roots converge to  $(n - m)$  straight-line asymptotes intersecting at: 
$$c = \frac{\sum_{k=1}^n p_k - \sum_{i=1}^m z_i}{n - m} \in \mathbb{R}$$
- If  $(\beta > 0)$ , the asymptotes intersect the real axis at angles 
$$\phi_k = \frac{\pi + k\pi}{n - m}, \quad k = 0, \dots, n - m - 1$$

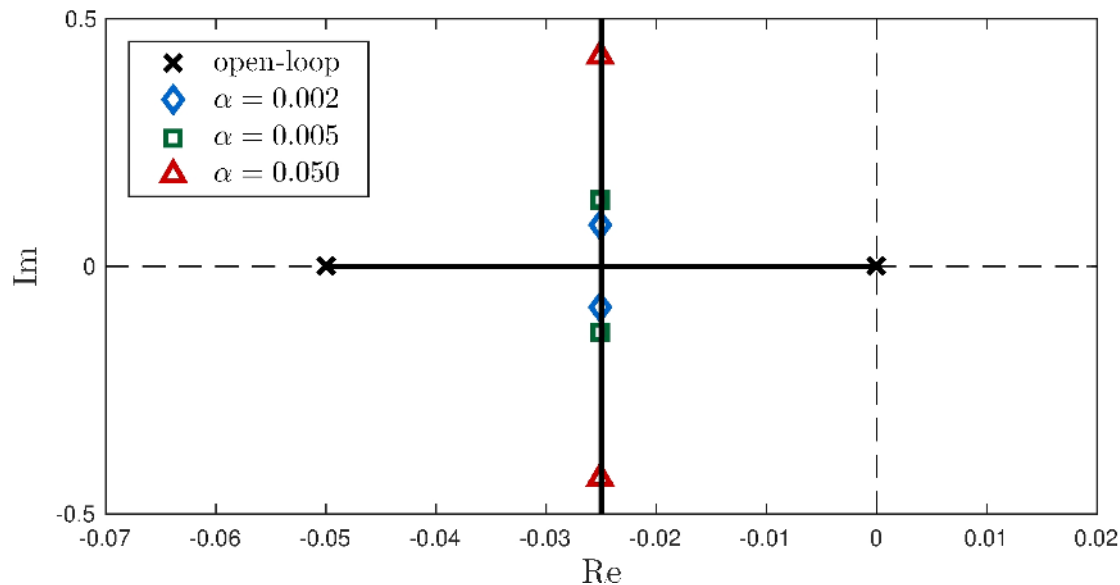
# Root-locus rules

## Example: car model under integral feedback

$$\alpha = K_i, \quad L(s) = G(s), \quad C(s) = \frac{3.7}{s(s + 0.05)}$$

### Asymptotes

$$c = \frac{-0.05 + 0}{2} = -0.025, \quad \phi_1 = \frac{\pi}{2}, \quad \phi_2 = \frac{3\pi}{2}$$



# Root-locus rules

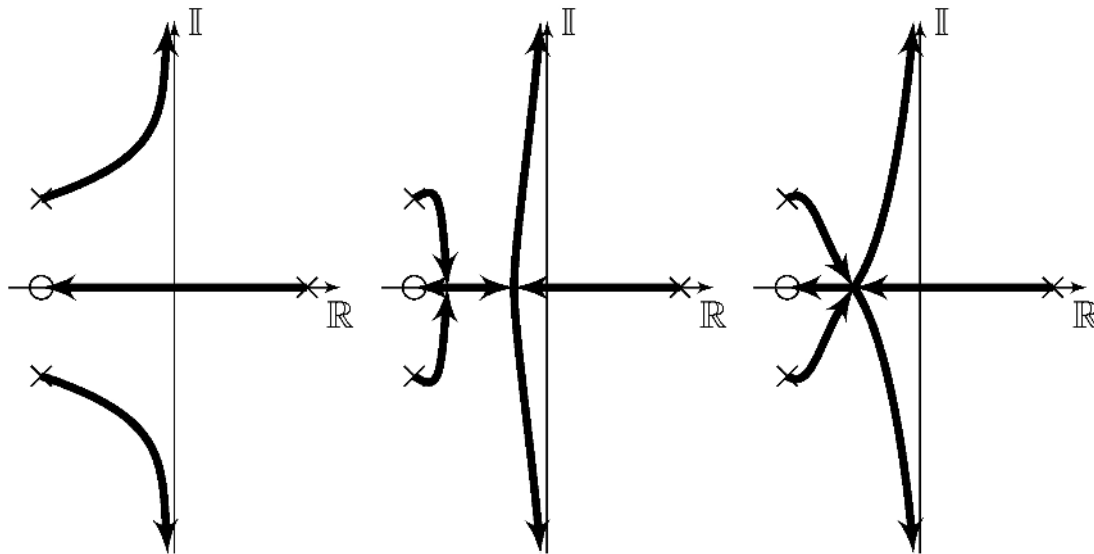
Example: model with three poles and one zero

$$\begin{aligned} p_1 &= a, & p_2 &= -a + j b, & p_3 &= -a - j b, & z_1 &= -a \end{aligned}$$

2 asymptotes

$$\begin{aligned} \sigma &= \frac{(-a - j b) + (-a + j b) + (a) - (-a)}{3 - 1} = 0, & \phi_1 &= \frac{\pi}{2}, & \phi_2 &= \frac{3\pi}{2} \end{aligned}$$

Possible root-loci



# Root-locus rules

## Example: car model under proportional-integral feedback

$$\begin{aligned} G &= \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}, & K(s) &= K_p + \frac{K_i}{s} = K_p \frac{s + K_i/K_p}{s} \end{aligned}$$

## Loop transfer-function

$$\begin{aligned} \alpha &= K_p, & L &= G, & C &= \frac{3.7 \left( s + \frac{K_i}{K_p} \right)}{s(s + 0.05)} \end{aligned}$$

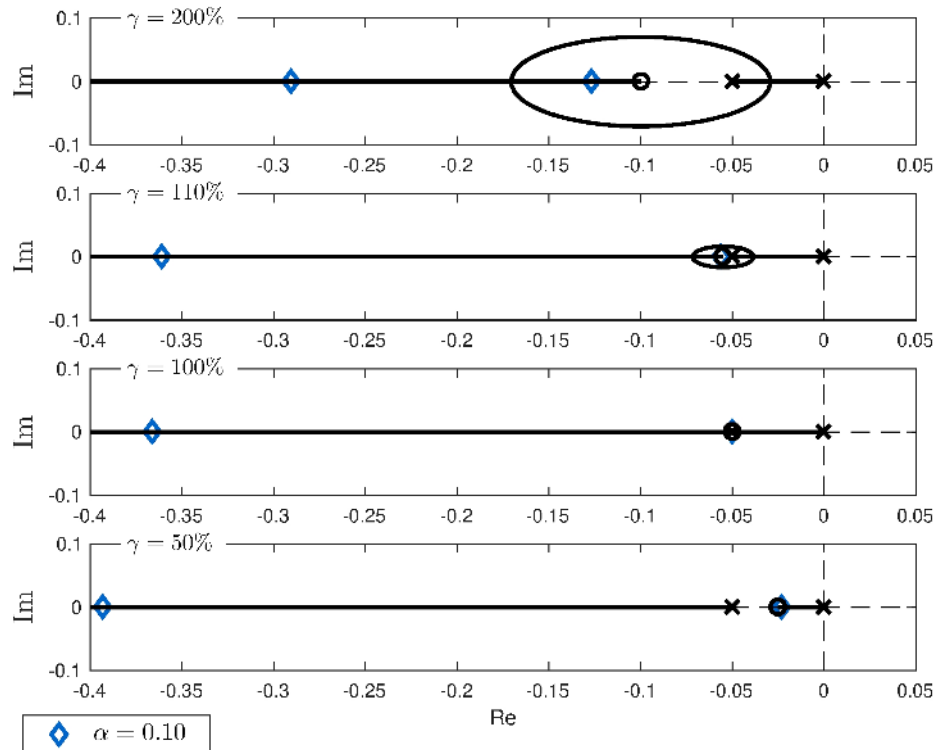
## Choice of controller zero

$$\frac{K_i}{K_p} = \gamma \frac{b}{m} = \gamma \cdot 0.05$$

- $\gamma = 1$  is a pole-zero cancellation

# Root-locus rules

Example: car model under proportional-integral feedback



# 6.5 Control of the Simple Pendulum - Part I

## Proportional-derivative feedback

### Controller

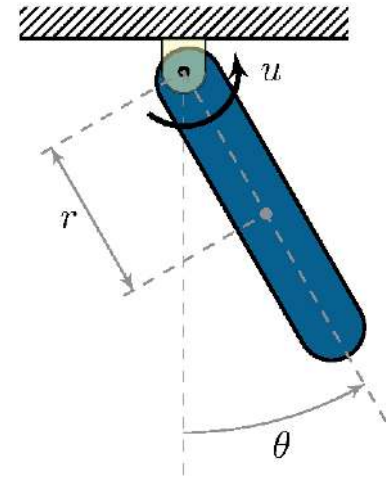
$$C_{\mathrm{PD}}(s) = K_d (s + z), \quad z = \frac{K_p}{K_d}$$

### Data

$$\begin{aligned} m &= 0.5 \text{ kg}, & \ell &= 0.3 \text{ m}, & r &= \ell/2 = 0.15 \text{ m} \\ b &= 0 \text{ kg/s}, & g &= 9.8 \text{ m/s}^2, & J &= \frac{m \ell^2}{3} = 0.015 \text{ kg m}^2 \\ J_{12} &= 3.75 \times 10^{-3} \text{ kg m}^2, & J_r &= J + m r^2 = 0.015 \end{aligned}$$

### Gains

$$K_d = 3, \quad m, \quad g, \quad r \approx 0.21, \quad K_p = 2 \sqrt{J_r m, g, r} - b \approx 2.21, \quad z \approx 10.5, \quad C_{\mathrm{PD}}(s) \approx 0.21 (s + 10.5)$$



# Proportional-derivative feedback

## System model

$$G_{\pi}(s) = \frac{66.7}{(s+7)(s-7)}$$

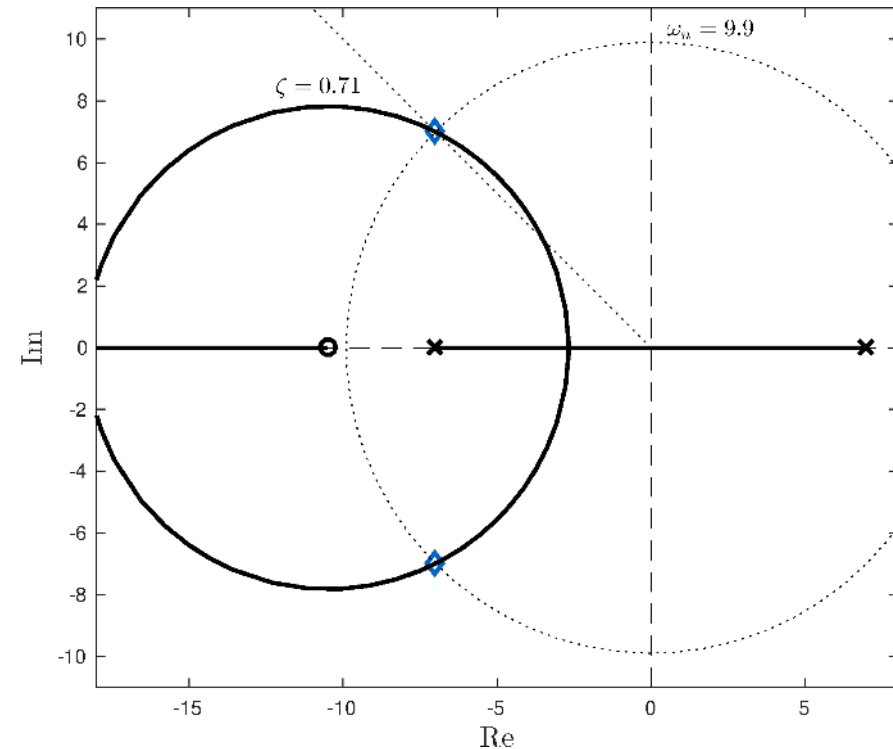
## Controller

$$K(s) = C_{\mathrm{PD}}(s) = K_d (s + z), \quad \text{and } z = \frac{K_p}{K_d} > 0$$

## Root-locus

$$\alpha = K_d, \quad L = \frac{1}{K_d} G_{\pi}, \quad C_{\mathrm{PD}} = \frac{66.7 (s + z)}{(s+7)(s-7)}$$

$$\omega_n \approx 10.5$$



- Diamonds:  $(K_d \approx 0.21 \Rightarrow \omega_n \approx 9.90, \zeta \approx 0.71)$
- PD controller is not proper, hence not realizable: we need a pole
- How about a pole at the origin for some desirable integral action?

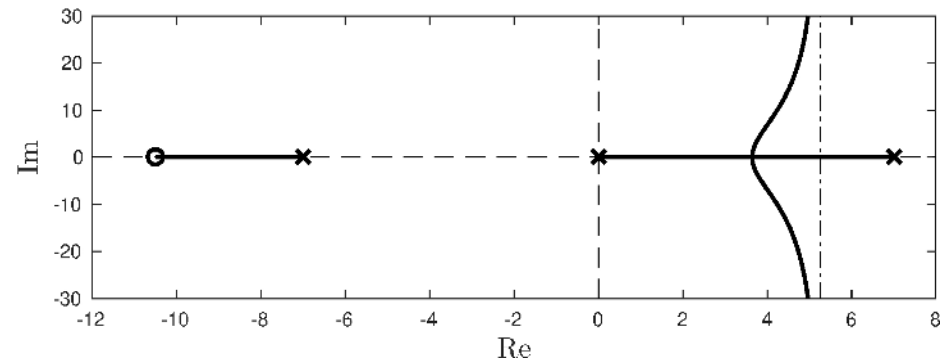
# Proportional-integral feedback

## Controller

$$K = K_p \frac{(s + z)}{s}$$

## Root-locus

$$\alpha = K_p, \quad L = \frac{66.7 (s + z)}{s (s + 7)(s - 7)}$$



- No  $(K_p > 0)$  can make closed-loop stable!



# Proportional-derivative feedback plus pole

## Controller

$$\begin{aligned} K(s) &= K \frac{s + z}{s + p} \end{aligned}$$

## Root-locus

$$\begin{aligned} \alpha &= K & L &= \frac{66.7 (s + z)}{(s + 7)(s - 7)(s + p)} \end{aligned}$$

## 2 asymptotes

$$\begin{aligned} c &= \frac{7 - 7 + (-p) - (-z)}{n - m} = \frac{z - p}{2}, & \phi_1 &= \frac{\pi}{2}, & \phi_2 &= \frac{3\pi}{2} \end{aligned}$$

## Stability

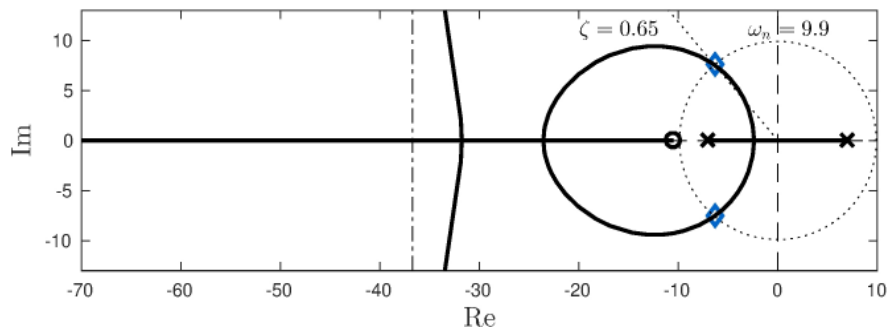
$$\text{Possible with large } (K) \text{ only if } \begin{aligned} c &< 0 \text{ \& \& implies } p &> z \end{aligned}$$

# Proportional-derivative feedback plus pole

## Root-locus

$$\alpha = K \quad L = \frac{66.7 (s + z)}{(s + 7)(s - 7)(s + p)}$$

## Choice of pole



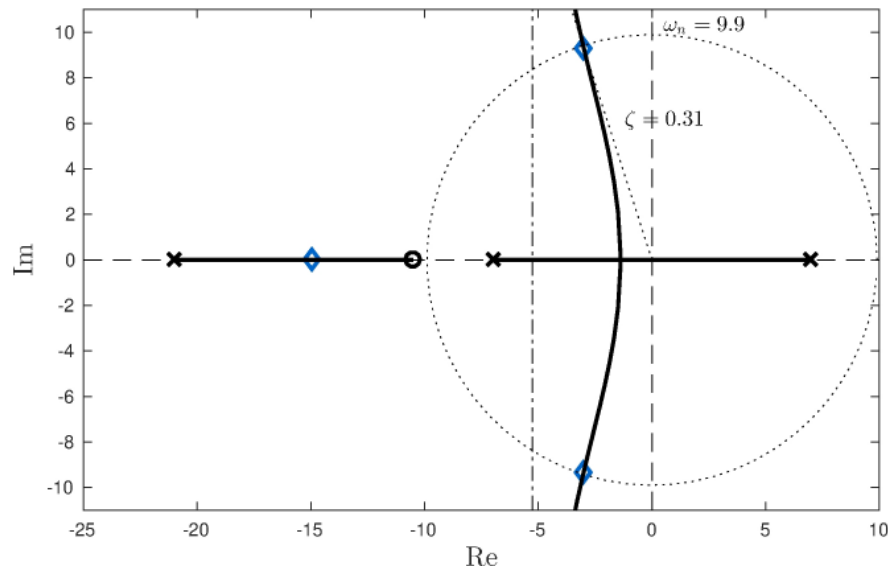
- $(p = 8 \text{ z } \approx 10.5)$
- Root-locus near origin is mostly unaffected

# Proportional-derivative feedback plus pole

## Root-locus

$$\alpha = K \quad L = \frac{66.7 (s + z)}{(s + 7)(s - 7)(s + p)}$$

## Choice of pole



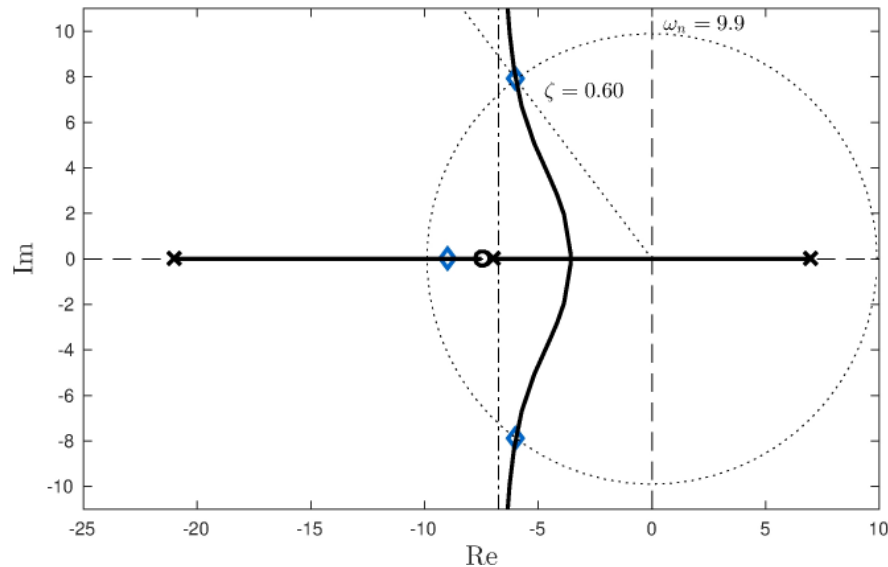
- $(p = 2z > z \approx 10.5)$
- Asymptote interferes with root-locus near the origin
- Low damping

# Proportional-derivative feedback plus pole

## Root-locus

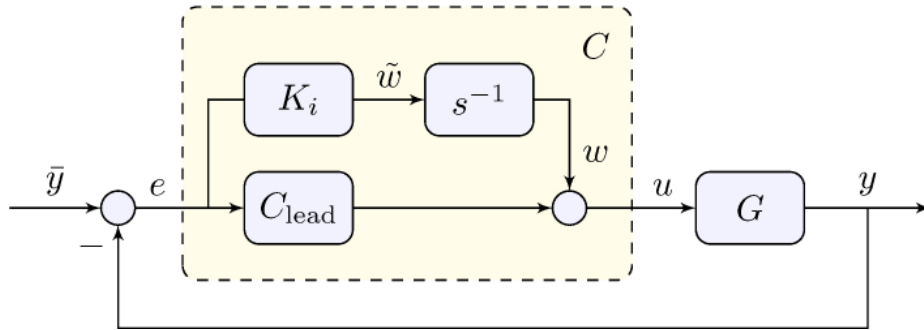
$$\alpha = K \text{ and } L = \frac{66.7 (s + z)}{(s + 7)(s - 7)(s + p)}$$

## Choice of pole and adjustment of zero



- $(p = 2z \approx 7.5)$
- Adjusting zero improves damping
- Final controller  $C_{\text{lead}}(s) \approx 3.83 \frac{s + 7.5}{s + 21}$

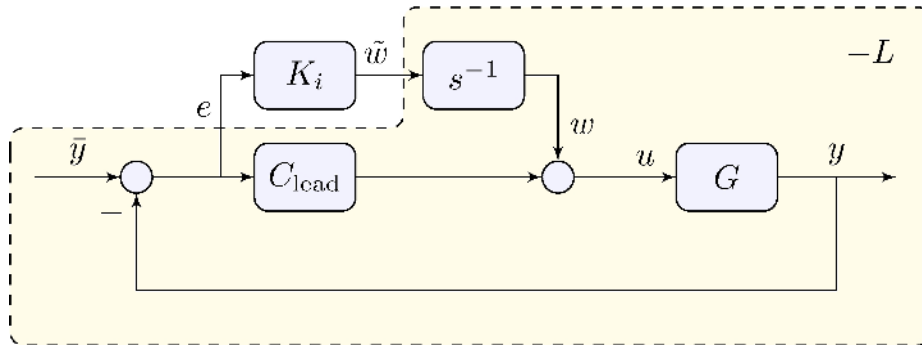
## Can we add integral action?



- Previous attempt failed to stabilize
- Additive approach means that stability is guaranteed for  $(K_i)$  small
- Structure of controller 
$$C(s) = \frac{K_i}{s} + C_{\text{lead}}(s) = K \frac{s^2 + (z + \frac{K_i}{K})s + p \frac{K_i}{K}}{(s + p)}$$
- What is  $(L)$ ?

# Can we add integral action?

## Root-locus analysis



## Loop transfer-function

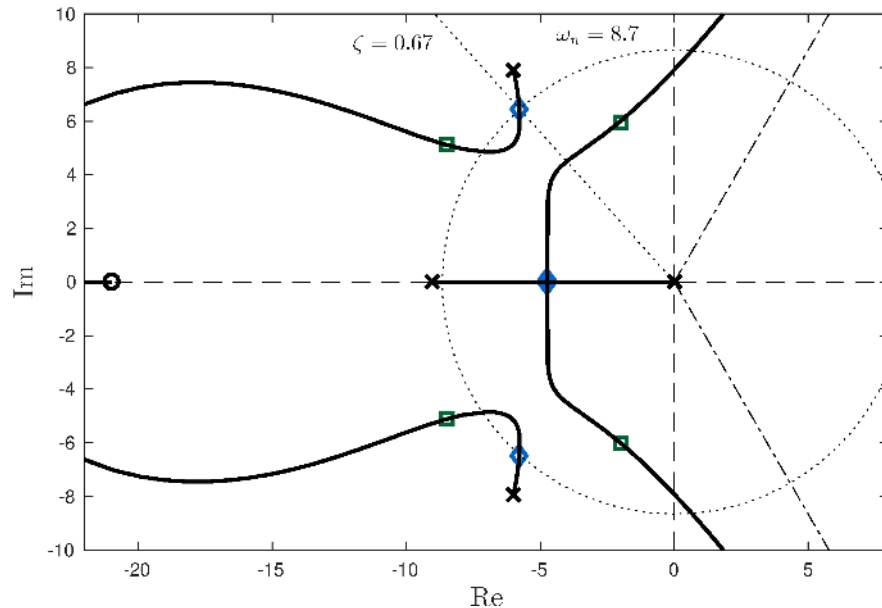
$$\begin{aligned} \alpha &= K_i, & L &= -\frac{E}{\tilde{W}} = -s^{-1} \frac{E}{W} = s^{-1} D = \frac{s^{-1} G_{\pi}}{1 + G_{\pi}}, C_{\text{lead}} \end{aligned}$$

# Can we add integral action?

## Root-locus analysis

$$\begin{aligned} \alpha &= K_i, & L &= -\frac{E}{\tilde{W}} = -s^{-1} \frac{E}{W} = s^{-1} D = \frac{s^{-1} G_{\pi}}{1 + G_{\pi}}, & C_{\mathrm{lead}} \end{aligned}$$

## Choice of integral gain



- Diamonds: double real poles,  $(\omega_n \approx 8.7)$
- Squares:  $(\omega_n \approx 9.9)$ , underdamped poles
- Final controller: 
$$C_{\mathrm{6}}(s) \approx 3.83 \frac{(s+6.86)(s+0.96)}{s(s+21)}$$
- See book for:
  - controller design around the stable equilibrium
  - direct root-locus design with integrator

# Learn more

- [Book website](#)



